MISB MOTION IMAGERY STANDARDS BOARD	MISB ST 1202.3
STANDARD	**02 March 2023**
Generalized Transformation Parameters	

1 Scope

This Standard (ST) describes a generalized method of transforming two-dimensional data (or points) from one coordinate system into a second two-dimensional coordinate system. This Generalized Transformation enables various image-to-image transformations such as an affine transformation by simply equating some parameters to be equal to zero. In addition, this Generalized Transformation may describe some homographic-like transformations.

This ST defines three items:

1) The different methods of implementation and constraints required to maintain certain transformation relationships.
2) The mandatory method for implementing uncertainty propagation on systems needing uncertainty information.
3) A KLV Local Set (LS) that represents all the parameters for the Generalized Transformation.

2 References

MISB references cited here-in reflect versions current to the publication date of a document. In the event of a MISB document correction, the corrected document will have a single letter Minor Version appended to the Major Version number per the MISB Document Development Process [1]. For example, corrections to ST 0001.2, which has a Major Version of 2, result in a new version titled ST 0001.2a, which includes a Minor Version of "a". The MISB will not update the referring document (this document) with the Minor Version number change. When acquiring any MISB reference listed below from an NGA repository the latest version may be either a Major or Minor Version.

[1] MISB Document Development Process 7.1, Dec 2021.
[2] MISB ST 0801.9 Photogrammetry Metadata for Motion Imagery, Oct 2021.
[3] MISB ST 1002.3 Range Motion Imagery, Mar 2023.
[4] MISB ST 1010.3 Generalized Standard Deviation and Correlation Coefficient Metadata, Oct 2016.
[5] MISB ST 1201.5 Floating Point to Integer Mapping, Jun 2021.
[6] MISB ST 0107.5 KLV Metadata in Motion Imagery, Oct 2021.
[7] MISB ST 0807.26 MISB KLV Metadata Registry, Jun 2022.

[8] NGA.STND.0017_3.0 Community Sensor Model (CSM) Technical Requirements Document (TRD) Version 3.9, Nov 2010.
[9] MISB ST 1107.4 Metric Geopositioning Metadata, Oct 2021.

3 Revision History

Revision	Date	Summary of Changes
ST 1202.3	03/02/2023	• Changed LS Document Version (Item 10) type to uint • Retired requirements 1 through 5 and 7 through 11 • Added Requirement -12 • Removed Universal Label's (keys) from LS Table 2 consistent with current MISB practice • Editorial changes to improve readability • Updated references

4 Acronyms and Definitions

CCFPA	Combined Composite Focal Plane Array
CPT	Child-Parent Transformation
CSM TRD	Community Sensor Model Technical Requirements Documents
CT	Chipping Transformation
DPIT	Default Pixel-Space to Image-Space Transformation
FLP	Floating Length Pack
FPA	Focal Plane Array
FSM	Fast Steering Mirror
KLV	Key Length Value
LS	Local Set
MISB	Motion Imagery Standards Board
NDT	No Defined Transformation
OT	Optical Transformation
ST	Standard

5 Introduction

This standard defines a Generalized Transformation based on the foundational two-dimensional projective transformation (also known as the eight parameters transform). The transformation data provides eight parameters for mapping between two-dimensional spaces. From the Generalized Transformation, this ST defines four types of commonly used (derived) transformations, and a methodology for extending support for additional derived transformations.

All derived transformations assume an unspecified parameter is equal to zero. This assumption helps in the implementation of transformations. As such, if all transformation parameters are equal to zero, then the resulting transformation returns an output identically equivalent to its input.

To prevent incorrect error propagation when invoking the stochastic model, it is necessary to account for all constraints that describe the individual transformation. Although not required, uncertainty information describing the quality of the transformation may accompany the transformation data. This ST describes the uncertainty information in the form of standard deviations and correlation coefficients.

To efficiently describe the Generalized Transformation and its uncertainty, this standard defines a Local Set (LS) to contain the transformation parameters and uncertainty.

This transformation must be associated with one of the two-dimensional spaces, which provides the context of the transformation. For its context, this LS requires a "parent" KLV set; thus, this LS functions when embedded in another LS.

6 Generalized Transformation

The Generalized Transformation describes a class of two-dimensional projective transformations intended for image-space coordinate transformations. The two-dimensional projective transformation is the foundation of the Generalized Transformation. This transformation defines a mathematical mapping from points on one plane to points on another plane and assumes a system of homogeneous coordinates. The following two equations provide a mathematical description of the plane-to-plane projective transformation relation of input to output image coordinates. The image coordinates may be in the form of decimal pixel coordinates or measured coordinates (i.e., measured in mm), see MISB ST 0809 [2] for information on image coordinate systems.

$$x_{out} = \frac{(1-A)x_{in} + By_{in} + C}{Gx_{in} + Hy_{in} + 1} \qquad \textbf{Equation 1}$$

$$y_{out} = \frac{Dx_{in} + (1-E)y_{in} + F}{Gx_{in} + Hy_{in} + 1} \qquad \textbf{Equation 2}$$

The form of Equation 1 and Equation 2 is slightly different than the description of a typical projective transformation, where the terms (1 - A) and (1 - E) are normally just A and E respectively. This modification allows for all values in the expression to be equal to zero without any need for special cases. With all constants (A through H) equal to zero, the transformation yields coordinate values identically equal to the input. This permits execution of the transformation regardless of the input data (e.g., if one or more of the parameters are zero).

As this transformation is a projective transformation, the inverse is a function of the original parameters. Thus, defining the forward transformation ***and*** the inverse transformation requires only one set of parameters. The inverse of Equation 1 and Equation 2, derived through a series of algebraic steps, results in Equation 3 and Equation 4 respectively.

$$x_{in} = \frac{\big((1-E) - FH\big)x_{out} + (CH - B)y_{out} + \big(BF - C(1-E)\big)}{\big(DH - G(1-E)\big)x_{out} + \big(GB - H(1-A)\big)y_{out} + \big((1-A)(1-E) - DB\big)} \qquad \textbf{Equation 3}$$

$$y_{in} = \frac{(GF - D)x_{out} + \big((1 - A) - CG\big)y_{out} + \big(DC - F(1 - A)\big)}{\big(DH - G(1 - E)\big)x_{out} + \big(GB - H(1 - A)\big)y_{out} + \big((1 - A)(1 - E) - DB\big)} \qquad \textbf{Equation 4}$$

Equations 1-4 define two-dimensional projective transformations and their inverses. In many imagery applications, a two-dimensional affine transformation requires six parameters. This set of six parameters is a subset of the original eight parameters described above. Section 7 contains various formulations of projective transformations and the constraints needed to create various "standard" transformations.

6.1 Transformation Types

Table 1 lists the derived transformations defined in this document along with corresponding enumeration values used in the subsequent Generalized Transformation LS.

Table 1: Derived Transformations List

Enumeration Value	Description	Units
0	Other – No Defined Transformation (NDT)	None
1	Chipping Transformation (CT)	pixels
2	Child-Parent Transformation (CPT)	pixels
3	Default Pixel-Space to Image-Space Transformation (DPIT)	mm
4	Optical Transformation (OT)	mm

6.1.1 Other – No Defined Transformation (NDT)

An enumeration value equal to "0" indicates an undefined transformation type. This transformation type provides for research and development purposes only and systems do not use this transformation type for production systems.

6.1.2 Chipping Transformation (CT)

An enumeration value equal to one "1" signifies the transmitted image is a chip (or sub-region) from a larger image. Examples of a chipped image are: 1) a digitally enlarged (zoom) sub-region of an image; 2) a sub-region of an image selected to reduce delivery bandwidth, or to provide an image of higher quality within the sub-region. Section 7.1.1 provides additional information on this transformation.

6.1.3 Child-Parent Transformation (CPT)

An enumeration value equal to two "2" indicates the transformation of a child focal plane array (FPA) to its parent FPA (e.g., see the example in MISB ST 1002 [3]). This CPT is a plane-to-plane transformation to transform between FPA's in image space. Section 7.1.2 provides additional information on this transformation.

6.1.4 Default Pixel-Space to Image-Space Transformation (DPIT)

An enumeration value equal to three "3" is the default pixel-space to image-space transformation. Section 7.1.3 provides additional information on this transformation.

6.1.5 Optical Transformation (OT)

An enumeration value equal to four "4" indicates the pixel data of an image is a translation, rotation, scale or skew from the originating FPA to the final optical focal plane. This may occur when the originating FPA is a subset of an entire optical focal plane. An example is a Combined Composite Focal Plane Array (CCFPA) sensor, where multiple focal plane array detectors combine to image a single optical focal plane. This optical transformation is a plane-to-plane transformation to transform from the FPA to the optical image plane.

In addition to providing a transformation from FPA to CCFPA, the optical transformation may also support the effects of coudé paths or Fast Steering Mirrors (FSM). Coudé path and FSM effects may mimic that of the transformation between FPA and CCFPA. They may also differ, however, by translating, rotating, scaling, or skewing the optical image plane. This document does not provide a description of how to model coudé path and FSM; however, modeling the effects of coudé path and FSM is possible using analytics. When systems determine the model parameters the OT provides the means to transmit them. Section 7.1.4 provides additional information on this transformation.

6.1.6 Extensibility for New Transformations

This standard is extensible allowing for new transformations to support new capabilities. The MISB will add the transformation with an assigned enumeration to Table 1 once approved.

6.2 Uncertainty Propagation

In many applications, the knowledge of the uncertainty of all estimated values is critical to understanding the performance of a system. Thus, it is desirable to provide a means to propagate the uncertainty information of the transformation parameters. The Generalized Transformation LS utilizes the format described in MISB ST 1010 [4] for transmitting standard deviation and correlation coefficient information.

This standard uses Parse Control Mode 2 of ST 1010, to support the most flexibility of standard deviation correlation coefficients. ST 1202 does not define the min or max values for any of the standard deviations, therefore the standard deviation format uses IEEE floating point. Parse Control Mode 2 allows for standard deviation sizes up to eight bytes. To save space, the MISB recommends using IMAPB (MISB ST 1201 [5]) when providing correlation coefficients in ST 1010. To reduce the size of the SDCC pack, applications should minimize the size of the standard deviation and correlation values appropriately, while maintaining the desired accuracy.

Requirement	
ST 1202.3-12	Where including Standard Deviation and Correlation Coefficients (Item 9) in the Generalized Transformation LS, the metadata producer shall use MISB ST 1010 Parse Control Mode 2.

The Standard Deviation Correlation Coefficient FLP, as defined in ST 1010, requires the parent LS (e.g., the Generalized Transformation LS in this case) to define the Source List. The Source List for ST 1202 is Items 1-8.

The projective transformation is the general case of the two-dimensional-to-two-dimensional transformation, and no constraints exist on the uncertainty propagation. See Section 7 for information on how to handle the uncertainty propagation for other transformations.

6.3 Concatenation of Transformations

For a single image, producers may provide up to four individual transformations to apply to the image. Each transformation will have a unique type and may provide its own uncertainty information. A benefit of projective transformations is that a combination of projective transformations is itself a projective transformation; however, the order in performing these transformations is critical. When producers provide two or more transformations, the receiver applies these transformations in the proper order. In the case of determining these transformations for sensor modeling purposes, which assumes an image-to-ground sequence, the defined order is as follows:

1. First transformation: chipping or digital zoom transformation. This transformation transforms the image pixel coordinates of the chipped or digitally zoomed image into the original image pixel coordinate system. Section 7.1.1 describes this transformation.
2. Second transformation: Child-Parent transformation. This transformation transforms the original image pixel coordinates (step 1 above) of the child image into the image pixel coordinate system of a parent image. Section 7.1.2 describes this transformation.
3. Third transformation: default pixel-space to image-space transformation. This transforms the pixel coordinates into units of millimeters (mm) and moves the origin to the center of the image. Section 7.1.3 describes this transformation.
4. Fourth transformation: image-space coordinates imaged on the focal plane into the optical image-space coordinate system. Section 7.1.4 describes this transformation.

The ground-to-image projection sequence is the inverse of the image-to-ground sequence. Uncertainty information may accompany all the above transformations.

6.4 Generalized Transformation Local Set

MISB ST 0107 [6] provides a set of baseline requirements for applying KLV metadata in Motion Imagery. The Generalized Transformation LS has the following requirements:

Requirement	
ST 1202.1-06	All metadata shall comply with MISB ST 0107.

ST 1202 is not a standalone Local Set; as such it receives its transformation context from its parent set.

The Generalized Transformation Local Set 16-Byte Universal Label registered in MISB ST 0807 [7] is:

06.0E.2B.34.02.0B.01.01.0E.01.03.05.05.00.00.00 (CRC 40498)

Table 2 summarizes the Generalized Transformation Local Set with the following column definitions:

- **Tag** indicates the Tag to use when specifying the item in the Composite Imaging Local Set
- **Name** is the name of the dictionary item
- **M/O** indicates if the Local Set item is Mandatory (M) or Optional (O)
- **Default** is the default value for the item when not specified
- **Notes** equates variables in the Section 6 equations or other information
- **Type** specifies the data type (binary format) when encoding the value into binary, see ST 0107. Note: The generalized type uint allows the application to define the number of bytes needed for a values representation, where the "Length" of the KLV triplet indicates a values length.

Table 2: Generalized Transformation Local Set

Tag	Name	M/O	Default	Notes	Type
1	x Equation Numerator – x factor	O	0 (Zero)	A in Equation 1	float
2	x Equation Numerator – y factor	O	0 (Zero)	B in Equation 1	float
3	x Equation Numerator - Constant factor	O	0 (Zero)	C in Equation 1	float
4	y Equation Numerator – x factor	O	0 (Zero)	D in Equation 2	float
5	y Equation Numerator – y factor	O	0 (Zero)	E in Equation 2	float
6	y Equation Numerator - Constant factor	O	0 (Zero)	F in Equation 2	float
7	Denominator – x factor	O	0 (Zero)	G in Equation 1 and Equation 2	float
8	Denominator – y factor	O	0 (Zero)	H in Equation 1 and Equation 2	float
9	Standard Deviation and Correlation Coefficients	O	N/A	Defined in MISB ST 1010	SDCC Pack
10	Document Version	M	N/A	Version of ST 1202 used	uint
11	Transformation Enumeration	O	0 (Zero)	Enumeration (see Table 1)	uint

7 Details

This section provides additional details on the mapping of the parameters in the unique transformation types, their inverses, and the uncertainty propagation represented within this ST.

7.1 Generalized Transformation LS

The Generalized Transformation Local Set supports transformation types when needed in the development of a sensor model. When receiving multiple transformations LS, a specific ordering of these transformations is mandatory per Section 6.3.

The following four subsections describe these transformations.

7.1.1 Chipping Transformation (CT)

The Chipping Transformation (CT enumeration value = 1) enables both image chipping and a special subset of image chipping known as digital zoom done within the Pixel Coordinate System defined by the Community Sensor Model (CSM) Technical Requirements Document (TRD) [8] (e.g., line and sample (or row and column) measured from the upper left-hand corner). Figure 1 and Figure 2 show examples.

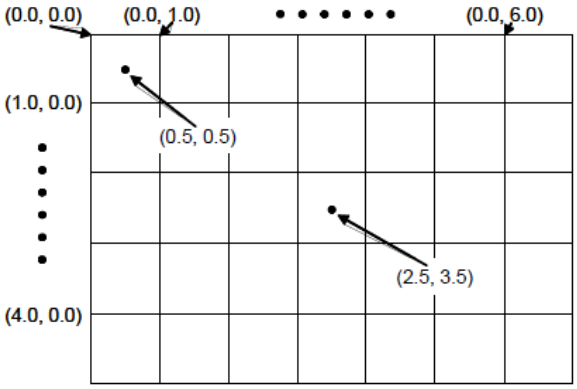


Figure 1: Pixel Coordinate System per CSM TRD

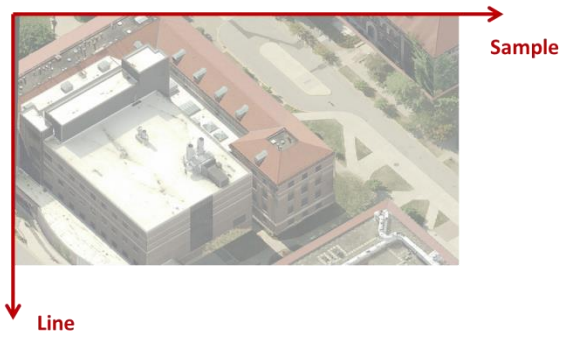


Figure 2: Pixel Coordinate System in an Image

Equation 5 gives the general form of the chipping transformation. A chipped or zoomed image is a sub-region of a larger image without rotation, as illustrated in Figure 3. The transformations needed for executing a sensor model must transform the chipped image pixel coordinates into the original image pixel coordinate space (i.e., the location of the original pixels must be known).

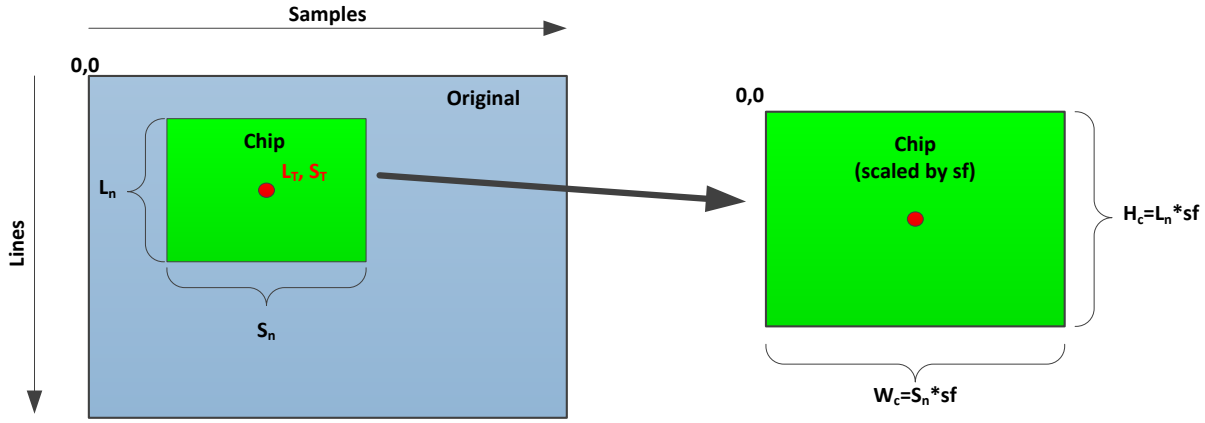


Figure 3: Example Chipping Transformation

$$\begin{bmatrix} L_O \\ S_O \end{bmatrix} = \begin{bmatrix} \frac{1}{sf} & 0 \\ 0 & \frac{1}{sf} \end{bmatrix} \begin{bmatrix} L_C \\ S_C \end{bmatrix} + \begin{bmatrix} L_T - \left(\frac{1}{sf}\right)\frac{H_C}{2} \\ S_T - \left(\frac{1}{sf}\right)\frac{W_C}{2} \end{bmatrix} \quad \textbf{Equation 5}$$

The equations below show how to compute the transformation parameters for chipping.

$$A = 1 - \frac{1}{sf} \quad \textbf{Equation 6}$$

$$B = 0 \quad \textbf{Equation 7}$$

$$C = L_T - (1 - A)\frac{H_C}{2} \quad \textbf{Equation 8}$$

$$D = 0 \quad \textbf{Equation 9}$$

$$E = 1 - \frac{1}{sf} \quad \textbf{Equation 10}$$

$$F = S_T - (1 - E)\frac{W_C}{2} \quad \textbf{Equation 11}$$

$$G = 0 \quad \textbf{Equation 12}$$

$$H = 0 \quad \textbf{Equation 13}$$

The translation values, L_T and S_T, in Equation 8 and Equation 11 respectively describe the location of the center of the chipped image within the original image. The value sf is the scale factor used to scale the image equally applied to both a line and sample. The variables L and S describe the line and sample coordinates, respectively, of the point of interest. In Equation 5, the

subscript O refers to the original image pixel coordinates and the subscript C refers to the chipped image pixel coordinates. Finally, the variables H_C and W_C are the chipped image height and width, respectively.

A special case of the chipping transformation is the Digital Zoom Transformation of the original image. A digital zoom maps the center region of the original image to a new image with new coordinates but the same dimensions as the original image, as illustrated in Figure 4. For this special case use the size of the original image to compute the last terms of Equation 5 as shown in Equation 14.

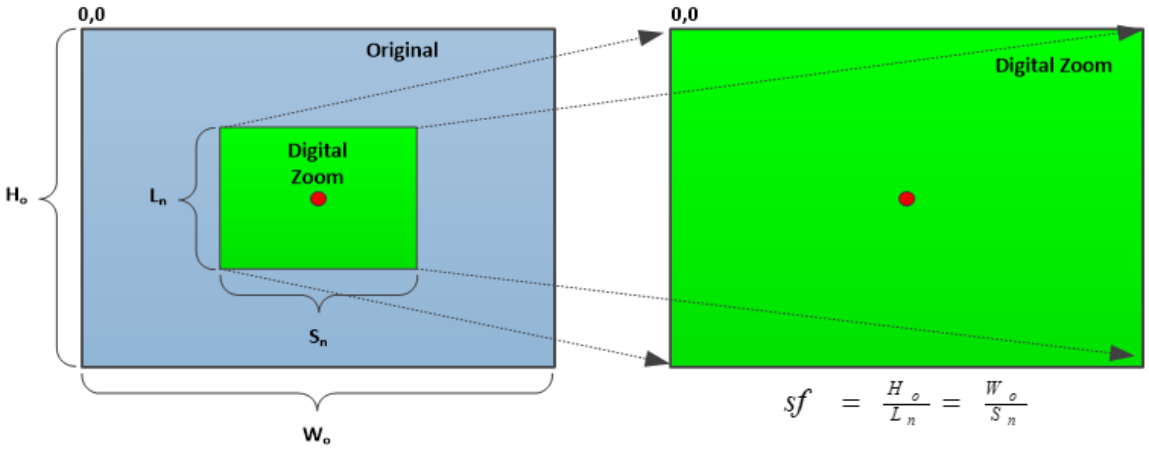


Figure 4: Digital Zoom Transformation

$$\begin{bmatrix} L_O \\ S_O \end{bmatrix} = \begin{bmatrix} \frac{1}{sf} & 0 \\ 0 & \frac{1}{sf} \end{bmatrix} \begin{bmatrix} L_C \\ S_C \end{bmatrix} + \begin{bmatrix} \frac{H_O}{2}\left(1 - \frac{1}{sf}\right) \\ \frac{W_O}{2}\left(1 - \frac{1}{sf}\right) \end{bmatrix} \qquad \textbf{Equation 14}$$

The equations below show how to compute the transformation parameters for digital zoom.

$$A = 1 - \frac{1}{sf} \qquad \textbf{Equation 15}$$

$$B = 0 \qquad \textbf{Equation 16}$$

$$C = \frac{H_O}{2}\left(1 - \frac{1}{sf}\right) \qquad \textbf{Equation 17}$$

$$D = 0 \qquad \textbf{Equation 18}$$

$$E = 1 - \frac{1}{sf} \qquad \textbf{Equation 19}$$

$$F = \frac{W_O}{2}\left(1 - \frac{1}{sf}\right) \quad \textbf{Equation 20}$$

$$G = 0 \quad \textbf{Equation 21}$$

$$H = 0 \quad \textbf{Equation 22}$$

The value sf is the scale value used to apply a digital zoom to an image. For example, for a 2X digital zoom sf = 2 equally applied to both line and sample. The variables L and S describe the line and sample coordinates, respectively, of the point of interest. The subscript O is in reference to the original image pixel coordinates and the subscript C is in reference to the chipped image pixel coordinates. Finally, the variables H_O and W_O are the original image height and width, respectively.

The chipping transformation only produces rescaled and translated images. Assuming the parameters for the chipping transformation are known, there is no need for uncertainty information about these parameters. Because of this, there is typically no stochastic model that accompanies this transformation.

The values defined in Equation 15 through Equation 22 or Equation 6 through Equation 13 can be used to define the inverse transformation using Equation 3 and Equation 4.

7.1.2 Child-Parent Transformation (CPT)

The Child-Parent Transformation (CPT enumeration value = 2) transforms a child focal plane array to its parent focal plane array. These two arrays are related within multiple sensors. An example of this is a co-boresighted sensor system with sensors contained within the same turret. In this formulation, one focal plane acts as the "parent" focal plane. This focal plane is what metadata, such as photogrammetry metadata, is in reference. The "child" focal plane is the image transformed into the "parent" sensor's coordinate system. The transformation can include rotation, translation, and scaling. This is an eight-parameter transformation via the General Transformation described in Equation 1 and Equation 2, where:

- The "*in*" subscripts represent the child image pixel coordinates.
- The "*out*" subscripts represent the parent image pixel coordinates.
- The variables L_{in} and S_{in} are the line and sample pixel coordinates in the child image.
- The variables L_{out} and S_{out} are the line and sample pixel coordinates in the parent image.

$$L_{out} = \frac{(1 - A)L_{in} + BS_{in} + C}{GL_{in} + HS_{in} + 1} \quad \textbf{Equation 23}$$

$$S_{out} = \frac{DL_{in} + (1 - E)S_{in} + F}{GL_{in} + HS_{in} + 1} \quad \textbf{Equation 24}$$

The CPT does not require any unique mapping into the metadata stream.

The transformation values in Equation 23 and Equation 24 can be used to define the inverse transformation in Equation 3 and Equation 4.

7.1.3 Default Pixel-Space to Image-Space Transformation (DPIT)

The Default Pixel-Space to Image-Space Transformation has two representations. The first is the CSM TRD defined approach for Motion Imagery; the second is a generalized approach for other imagery modalities. The definitions for these cases are in Sections 7.1.3.1 and 7.1.3.2 respectively.

7.1.3.1 CSM TRD Default Pixel to Image-Space Transformation

For Motion Imagery, the default transformation (DPIT enumeration value = 3) is the assumed transformation in constructing a CSM compliant sensor model of a ***full*** image. That is, the full focal plane array is the data transmitted and represented by the metadata stream. This is the transformation assumed in MISB ST 1107 [9], which converts the pixel coordinates into the image measured coordinates for the sensor model. Equation 25 defines this transformation.

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 & p2m_x \\ -p2m_y & 0 \end{bmatrix} \begin{bmatrix} L - \frac{H}{2} \\ S - \frac{W}{2} \end{bmatrix} \qquad \textbf{Equation 25}$$

The values $p2m_x$ and $p2m_y$ are the dimensions given to each individual pixel. These pixels may, or may not, be square pixels. The variables L and S describe the line and sample pixel coordinates, respectively, of the point of interest. The variables H and W are the full image height and width, respectively. Finally, the variables x and y are the image measured coordinates.

Typically, ST 1107 would invoke this transformation as its parent LS; however, this transformation could embed into other parent LSs to invoke the default transformation. The following equations describe this transformation when present in the metadata stream:

$A = 1$	**Equation 26**
$B = p2m_x$	**Equation 27**
$C = -p2m_x \frac{W}{2}$	**Equation 28**
$D = -p2m_y$	**Equation 29**
$E = 1$	**Equation 30**
$F = p2m_y \frac{H}{2}$	**Equation 31**
$G = 0$	**Equation 32**
$H = 0$	**Equation 33**

As this is the default transformation and considered a known constant, there is typically not a stochastic model that accompanies this transformation.

The values defined in Equation 25 through Equation 33 may be used to define the inverse transformation using Equation 3 and Equation 4.

7.1.3.2 Generalized Pixel to Image-Space Transformation

For other imagery modalities not using the CSM TRD defined pixel-space to image-space transformation, the transformation (DPIT enumeration value = 3) is the assumed generalized transformation that defines the transformation between the pixel image coordinate system and the image coordinate system. Equation 34 and Equation 35 describe the transformation, with the image pixel coordinates represented by the "*in*" subscripts, and the image measured coordinates represented by the "*out*" subscripts. This transformation, defined previously, is:

$$x_{out} = \frac{(1-A)L_{in} + BS_{in} + C}{GL_{in} + HS_{in} + 1} \qquad \textbf{Equation 34}$$

$$y_{out} = \frac{DL_{in} + (1-E)S_{in} + F}{GL_{in} + HS_{in} + 1} \qquad \textbf{Equation 35}$$

The generalized default pixel-space to image-space transformation introduces no new variables. It uses the variables L_{in} and S_{in} to describe the line and sample coordinates in the pixel image and uses the variables x_{out} and y_{out} to describe the x and y coordinates in the image measured coordinate system. The transformation values in Equation 34 and Equation 35 may be used to define the inverse transformation in Equation 3 and Equation 4.

7.1.4 Optical Transformation (OT)

The Optical Transformation (OT enumeration value = 4) is the assumed transformation used in transforming a FPA to a CCFPA or the effects of coudé path and FSM. These two arrays are optically related within one sensor. The relationship occurs when the pixel data of the image is a translation, rotation, scale or skew from the optical focal plane. The OT transforms the focal plane array to the optical focal plane array. Like the CPT, the two arrays consist of an *"out"* and *"in"* array. The optical focal plane array is the *"out"* array while the originating FPA is the *"in"* array. The eight-parameter transformation via the General Transformation described in Equation 1 and Equation 2 repeats below for reference.

$$x_{out} = \frac{(1-A)x_{in} + By_{in} + C}{Gx_{in} + Hy_{in} + 1} \qquad \textbf{Equation 36}$$

$$y_{out} = \frac{Dx_{in} + (1-E)y_{in} + F}{Gx_{in} + Hy_{in} + 1} \qquad \textbf{Equation 37}$$

The OT introduces no new variables. It uses the variables x_{in} and y_{in} to describe the originating FPA image measured coordinates, and the variables x_{out} and y_{out} to describe the optical image measured coordinates.

The OT does not require any unique mapping into the metadata stream when embedded into a parent LS.

The OT above is the General Transformation described in the main body of this document. Uncertainty propagation for the OT is the same as the General Transformation - See Section 6.2.

The values defined in Equation 36 and Equation 37 may be used to define the inverse transformation in Equation 3 and Equation 4.

8 Retired Requirements

Requirement(s)	
ST 1202.1-01 (Retired)	Additional derived transformations supported by the MISB shall be added to MISB ST 1202 Table 1 along with supporting information regarding type and use.
ST 1202.1-02 (Retired)	When uncertainty information of the Generalized Transformation parameters is available, uncertainty information shall be represented by a Standard Deviation Correlation Coefficient Floating Length Pack (FLP) in accordance with MISB ST 1010.
ST 1202.1-03 (Retired)	The matrix size in the Standard Deviation Correlation Coefficient FLP shall be eight (8) to represent all the parameters in the Generalized Transformation.
ST 1202.1-04 (Retired)	The Standard Deviation Correlation Coefficient FLP LS shall order its entries for the eight elements of the Generalized Transformation LS in the same order as the first eight parameters of MISB ST 1202 - Table 2.
ST 1202.1-07 (Retired)	The Document Version (Item 10) of MISB ST 1202 utilized shall always be sent in the Generalized Transformation LS.
ST 1202.1-08 (Retired)	When the enumeration value corresponding to the transformation type is not populated in the Generalized Transformation LS, the value shall be assumed to be equal to zero (0) indicating No Defined Transformation.
ST 1202.1-09 (Retired)	The MISB ST 1202 Local Set shall be embedded within a LS that provides context for the transformation.
ST 1202.2-10 (Retired)	Standard deviation values shall be represented by four (4) byte floats.
ST 1202.2-11 (Retired)	Correlation coefficient values shall be mapped into two (2) byte integers using IMAPB(-1.0, 1.0, 2) (see MISB ST 1201).
ST 1202.1-05 (Retired)	Transformations shall be performed in the following order: 1) chipping, 2) child-parent, 3) default pixel-space to image-space and 4) image-space coordinates imaged on the focal plane into the optical image-space coordinate system.